



# **B737 NG THROTTLE QUADRANT SETUP MANUAL Ver: 3.3**

Revolution- Simproducts

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# INTRODUCTION

Dear Revolution-Simproducts Customer,

The Throttle Quadrant you have purchased is actually the result of a very long development process and a lot of hard work. We believe we can be proud of the result.

It is the best throttle to work with supported flight models in terms of functionality.

Every single function of the throttle is programmed based on the operations manuals written by the actual aircraft manufacturer. In addition to that we've had assistance and input from real Boeing 737 pilots.

However, some of the functionalities of the real throttle can not be simulated due to the limitations of either the flight simulator or the aircraft model used, we see these as minor differences which do not have any negative effect on the flying or the operation of the airplane.

In the different sections of this manual you will find the specific instructions for users and their respective setup methods.

Since flight simulation and cockpit building is a complex task and given that there are no real STANDARDS laid out, we are aware that different Aircraft Models and Avionic Packages will require a different setup. And then of course, each user will have a different setup and a different requirement for sending inputs into their Simulator.

Consequently, this manual will attempt to instruct the user in how to install and configure the Driver Program and also on the different methods of sending control inputs to your specific Aircraft or Avionics Package so that you have reliable and full functionality of your new TQ.

Basic setup will be the same for all flight models used. A good example of that will be the FSUIPC setup that we have gone into detail with in this document.

We are constantly developing Drivers for all the leading Add-On Aircraft and Avionics Packages, therefore this manual will constantly be updated with every new flight model implementation completed. With every new manual there will be the revision list for the points that have changed.

**We hope you will enjoy flying your airplane with our throttle.**



## REVISION LIST

| Version    | Description                 |
|------------|-----------------------------|
| 1.0 to 2.0 | Not Issued                  |
| 2.1        | Controlled Circulation Only |
| 2.2        | Controlled Circulation Only |
| 2.3        | Issued                      |
| 2.4        | Issued                      |
| 3.0        | Issued                      |
| 3.1        | Issued                      |
| 3.2        | Issued                      |
| 3.3        | Issued                      |

## Prerequisites

1. Your computer should have sufficient free USB slots for the TQ cards to be connected.

[Lite Version](#) TQ's will require 1 free USB slot for the Joystick Controller

[Motorised 737 White TQ](#) TQ's will require 2 free USB slots. One for the Controller and one for the Motor Controllers.

[Motorized 737 White PRO TQ's](#) will require 2 free USB slots.

[ALL TQ's](#) delivered starting from February 2011 (except Non Motorized ones) onwards will use only 2 USB connection. This both to save one USB port as well as easier installation of buttons and switches programming.

2. A Mains Power Source (110/230/240v) for the throttle power supply unit(s).

[Lite Version](#) TQ's do not require an external Mains Power Source

[Motorised 737 White TQ](#) TQ's are fitted with 1 Power Supply Unit

[Motorized 737 White PRO TQ's](#) are fitted with 1 Power Supply Unit

[ALL TQ's](#) delivered starting from February 2011 (except Non Motorized ones) will use only one power supply

3. Licensed copy of FSUIPC for the functionality setup. Please Update to current version
4. SIOC V3.52b (or later) installation files and your Script File.
5. Working copy of your Flight Simulator
6. Licensed copy of your Add on Aircraft or Avionics Package
7. For Customers with Backlighting Installed, a 12vdc Power Supply

## TQ Versions explained :

To prevent any misunderstanding from this point onward TQ's with two card will be numbered as Version 1.6, Version 1.7 etc.

Due to nature of continuous development process and based on customer questions we are progressing our TQ's with different configurations for an easier setup process.

Version 1.6 TQ's;

These are the TQ'S with one motor card an one USB card. The second card in these TQ's are not joystick cards therefore can not be programmed via FSUIPC. They have their own software to be downloaded and installed and in the relevant part of this manual it will be explained.

Version 1.7 TQ's;

These are TQ's again with one motor card an done TQ manager card as we call it. The only difference from the version 1.6 TQ's is the reverser axis is now controlled by SIOC and not the TQ manager program. This has been necessary with the introduction of Windows 7 since some of our customers had a difficulty on getting the axis read correctly by the TQ manager program. Therefore we have decided to have the control on SIOC as it has nothing to do with classical axis reading mechanisms of Windows.

**Version numbers of TQ's delivered after September 1st, 2011 will be written on the placards of TQ's.**  
Please read the manual and setup session accordingly.

## How to Setup your SIOC & Understanding How the Cards Are Numbered

You may run the SIOC program, either on your actual FS computer, or on a networked computer that is running Pete Dowson's Wideclient. Whichever computer you run SIOC on, this is the computer that you will attach your TQ's USB plugs to.

In the event that you do indeed run SIOC on a networked machine, it is required that WideFS is installed on your flightsim computer and that the Wideclient can connect to it over the network.

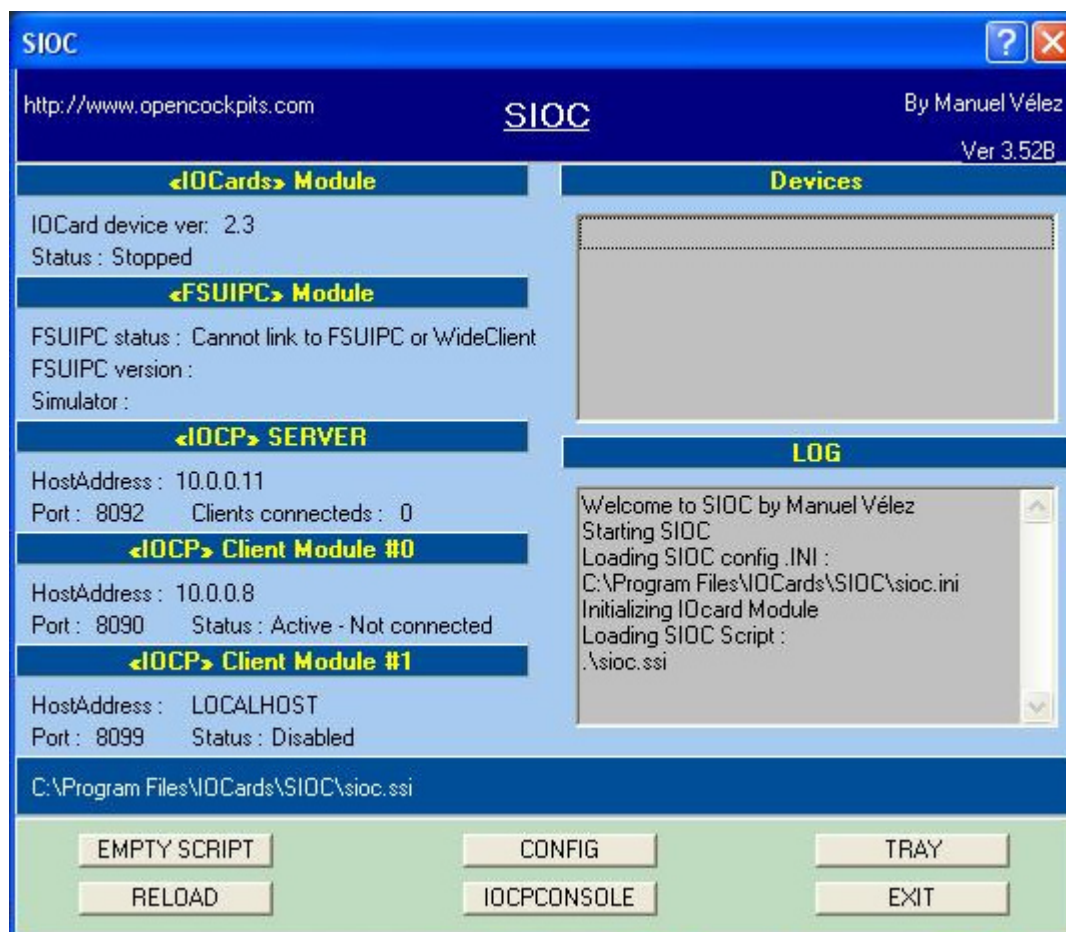
Detailed instructions on how to do that are in the WideFS documentation that is available on Peter's page at [www.shiratti.com/dowson](http://www.shiratti.com/dowson)

Owners of the Lite Version TQ may skip the next part and proceed direct to the 'Setting Up Your TQ in FSUIPC' section at Step 7.

**Step 1:** Install a copy of SIOC on whichever computer you decide to connect the TQ to. This must be v3.52b or above. You may get this from our downloads page

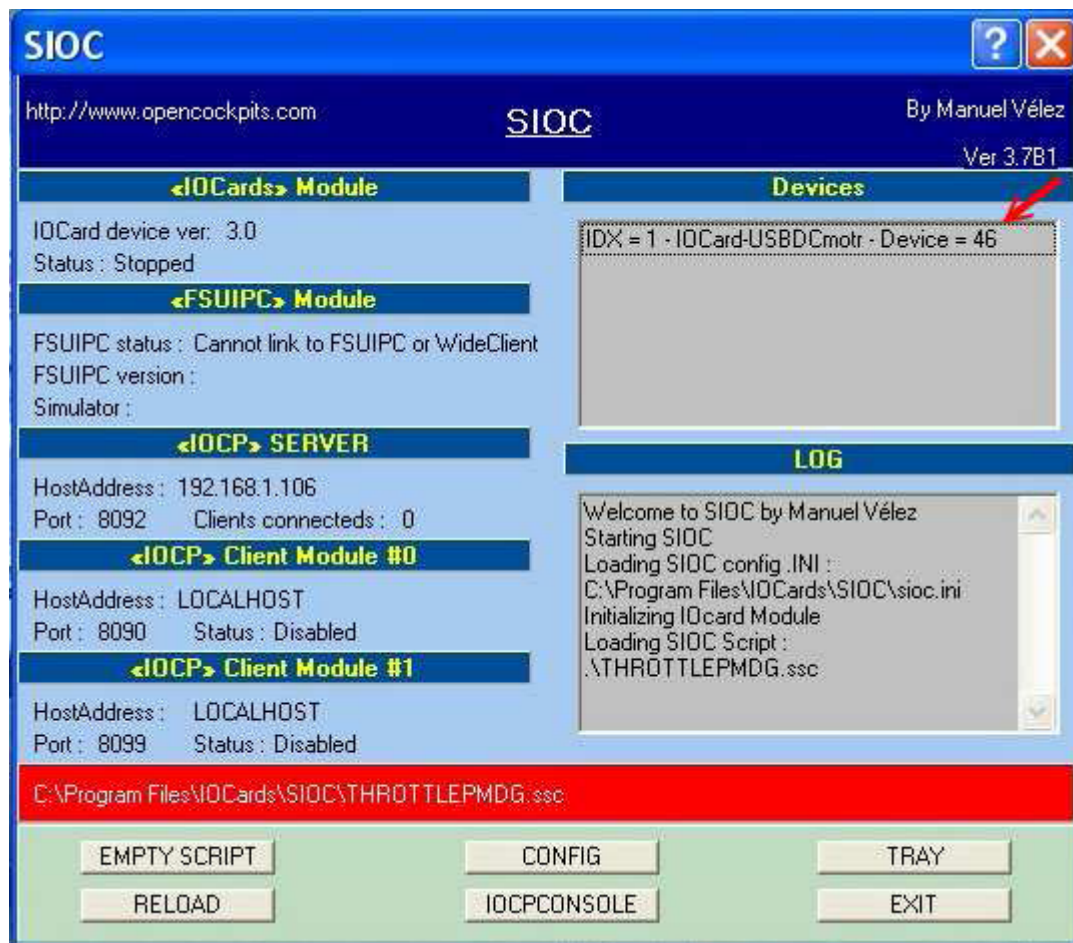
<http://www.revolution-simproducts.com/downloads.html> or visit <http://www.opencockpits.com>

**Step 2:** Once you have installed SIOC, navigate to the folder where you installed it and run SIOC.exe. You will see a window like this if the program has installed correctly.



Close the program and for ease of use, create a shortcut to sioe.exe on the desktop.

**Step 3:** Connect the first USB Lead (Card 1) to the computer and let it install. Re-start sioc.exe. SIOC will scan your system and if it detects any connected devices, it will assign a unique number to the device, in our case the USBDCMotr card like this.



Window should not be definitely identical. The red band should not necessarily be there so if you do not see it do not worry.

In the above example, look at the highlighted line in the Devices box. This reads :

IDX = 1 – IOCard-USBDCMotr – Device = 46

IDX = 1 is the index number that SIOC has assigned to the device.

If ones the card is connected IDX definiton comes with a \* sign this is perfectly OK. Once you alter the necessary lines in SIOC.ini file as explained below the indication on SIOC main screen will change and read the correct IDX number from the ini file.

**IMPORTANT:** When the card is identified for the first time and if IDX comes with \* sign, make a note of the device number for whichever card you have connected. If card connection is marked as CARD1 Or Motor Card 1 behind the throttle then you will change the ini file settings as explained below accordingly. IDX numbers are assigned by you not by the system. Only Device numbers are given by the system.

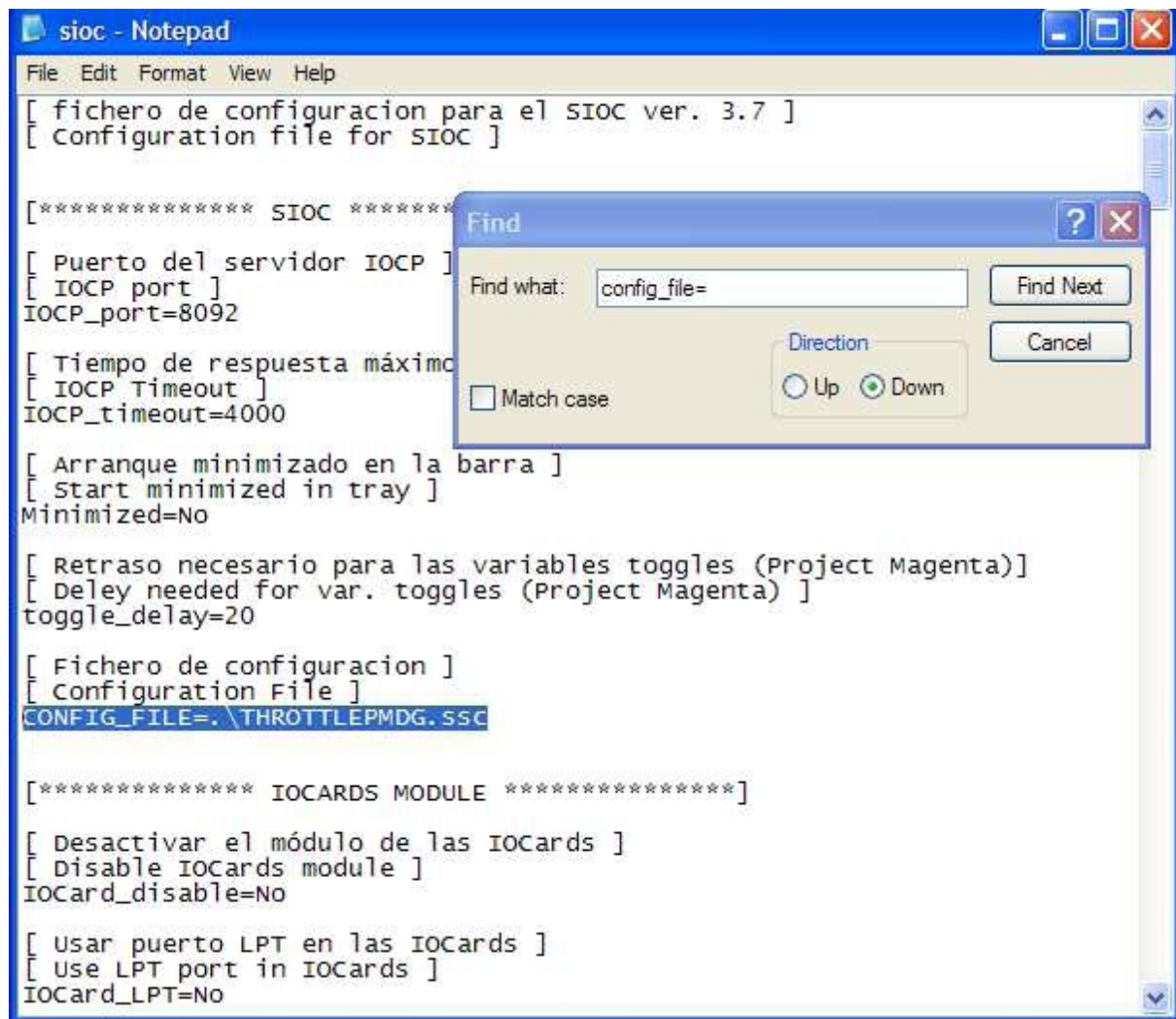
IOCard-USBDCTMotr is the device that SIOC has detected on the system. Device = 46 is the unique device number that SIOC has assigned to the DCMotr card. Please make a note of both the IDX and Device numbers as you will need these in the next section to setup the correct configuration of the controls in the SIOC.ini file. In our example, the numbers are IDX = 1 and Device=46. Your numbers will be different dependant on whether you have other IOCards on your system and of course SIOC will assign a unique device number to your card.

**Step 4 :** Now we have the IDX and Device numbers for our card or cards, it's time to tell SIOC how to communicate with the installed cards. We do this in the sioc.ini file that you can find in the folder you installed SIOC into, typically C:\Program Files\IO Cards\SIOC.

We will also need to tell SIOC which file to use to configure the TQ for use with your application. In our example, we are using the PMDG 737, so our file that does this is the THROTTLEPMDG.ssi script. You will have a differently named script depending on which TQ you have ordered, your application and it will include your name because we now provide each customer with a personalized script. Please see the section Identifying Your Script towards the end of this document.

Simply change the name you enter in the sioc.ini file to the one you wish to use. First copy the script file (in our example THROTTLEPMDG.ssi) we have supplied you with into the folder that contains your SIOC installation. This script will be downloaded from our web site download section.

Ok, open the sioc.ini file using a text editor like Notepad and using the Edit\Find Menu option, find the line containing CONFIG\_FILE=, like this :

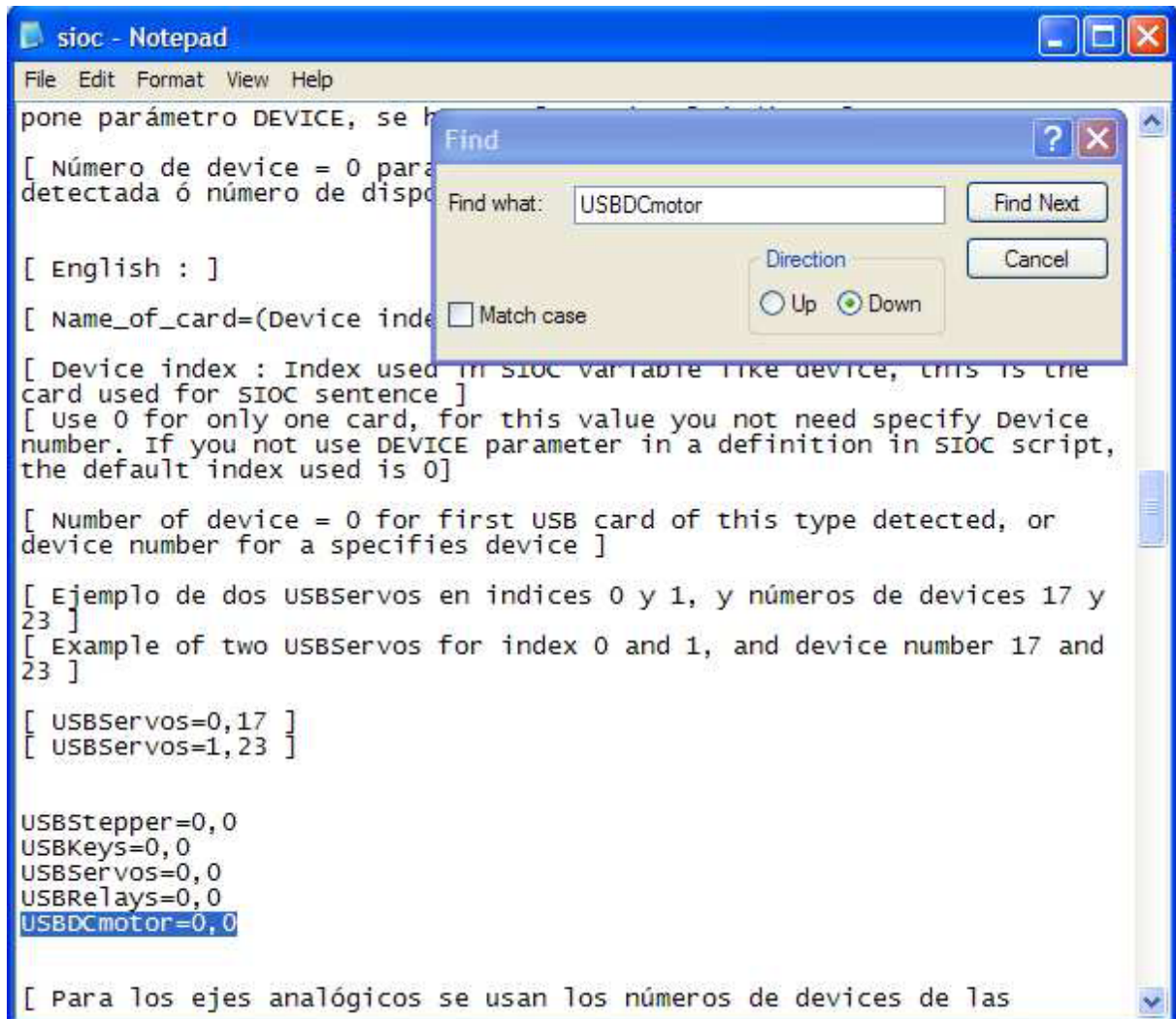


Write the name of your configuration file immediately after the slash. In our example it is the file we use for the PMDG, hence CONFIG\_FILE=.\THROTTLEPMDG.ssi

That has now told SIOC which file to read to allow us to communicate with the cards in the TQ.

What we now need to do is to tell SIOC which cards are controlling which functions and from which cards we will read input data.

Use the Edit\Find option to find the line containing the USBDCmotor= line like this :



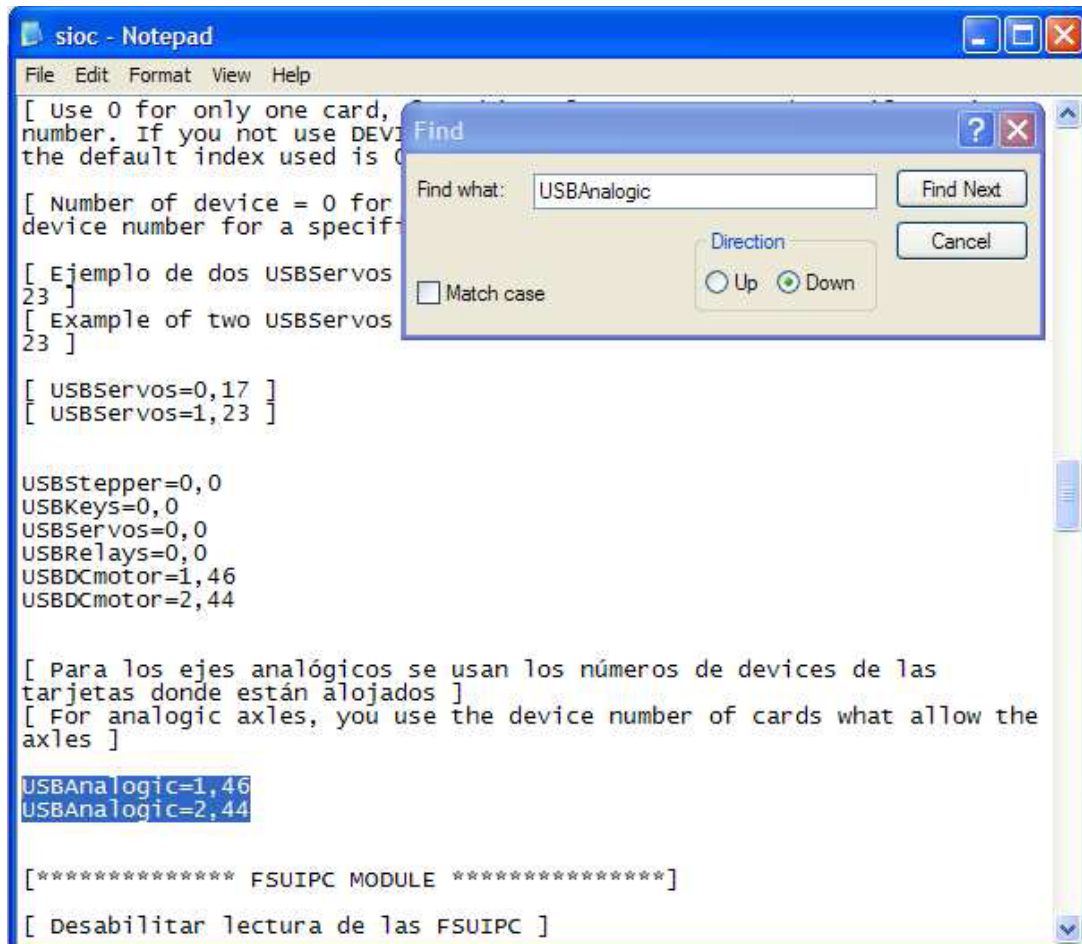
The line above reads USBDCmotor=0,0 . What we will do is change this to reflect the numbers of our card that SIOC has reported. In our example these were :

IDX = 1 – IOCard-USBDCMotr – Device = 46

We write the IDX number first and then the Device number second separated by a comma.

The line will then read :  
USBDCmotor=1,46

**Step 5 :** Ok, what we have done so far is to tell SIOC which script file to use to control the TQ and also tell it which card to use for what functions. But we also have to read input data from the cards as well for correct operation. This we do by telling SIOC from which card inputs from each of the potentiometers is read. The process is very similar to what we did with the USBDCmotor lines. But this time we will Find the lines which contain the USBAnalogic entries like this :



Simply edit the lines (or line) to contain the IDX and Device numbers that SIOC reported. Users of the standard Motorised TQ will only need to have one line (because they only have one card), but PROLINE TQ & Custom Function TQ customers will have to add a second line as in our example above because they of course have two cards. Save the file.

**IMPORTANT: IN A DEFAULT INSTALLATION OF SIOC THERE ARE SOME COMMENTED LINES THAT START WITH A '[' CHARACTER IN THE '.INI' FILE' . THESE ARE OK.**

**ANY LINE CONTAINED WITHIN THE '[' & ']' IS IGNORED. BUT, IF ANY LINE STARTING WITH ' MASTER X,X,X,X' IS THERE, DELETE IT. THIS CAUSES CONFLICT ON THE DEVICE NUMBERS AND THE THROTTLES DO NOT PERFORM.**

That's it, all done with editing the SIOC.ini file. Now in Step 7 we'll show you how to assign your buttons (and potentiometers if you have a standard TQ) in FSUIPC.

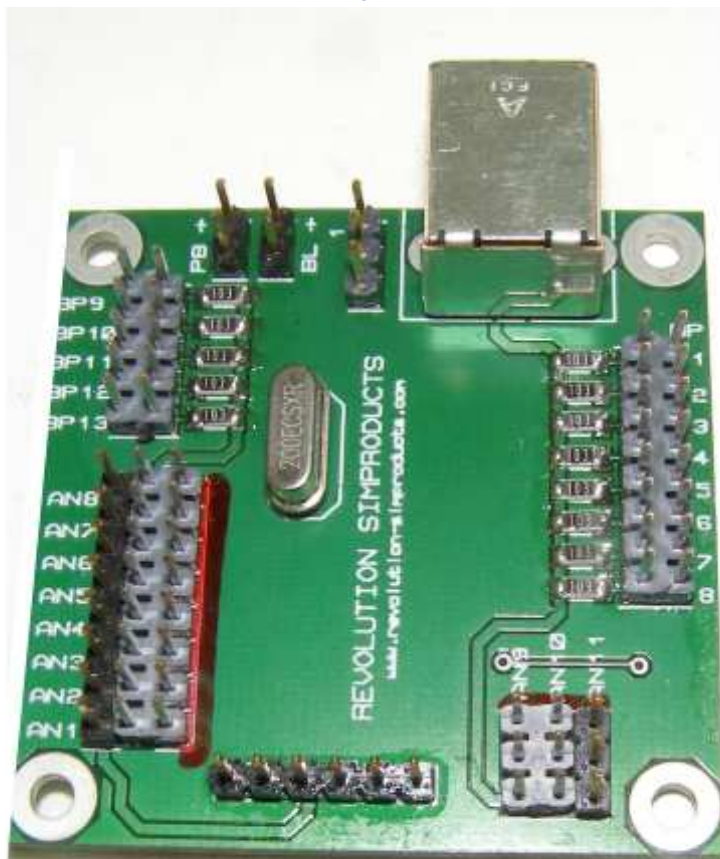
**Step 6 :** For Setting Up Your TQ in FSUIPC, you will need a registered version of the software. You may download it from [www.schiratti.com/dowson](http://www.schiratti.com/dowson)  
[VERSION 1.6 and above TQ owners please proceed to the last section marked REVOLUTION THROTTLE QUADRANT MANAGER](#)

Current versions at time of writing are v3.98 (FS9) and v4.60 (FSX)

First of all, we will deal with assigning the Buttons because all TQ customers will have this requirement.

\* NOTE Customer connecting their TQ to a networked computer over WideFS may also connect the Joystick Card USB lead to the same computer. FSUIPC is able to identify controllers attached to the networked machine when WideFS/Wideclient are connected and assign an identifier number to the controller for Button & Axis inputs.

### REVOLUTION THROTTLE QUADRANT MANAGER



We have developed this card to try to simplify the connections and calibration of our TQ. Your new TQ has been 'Factory Calibrated' and Tested. The card requires an external 12vdc power supply in order to function, control the Parking Brake Light and to manage the output to the Backlighting (if fitted). You will also require a spare USB socket on your PC system.

Please read this manual before attempting to install or calibrate your new TQ. It will help you understand the principles and method used to get ultimate performance from your new TQ.

Your new TQ will have been 'Factory Calibrated' and the values stored for a truly 'Plug & Play' Experience.

Once calibrated in house, your new TQ will have been Factory Tested to check operation and calibration of all Axes & Functions.

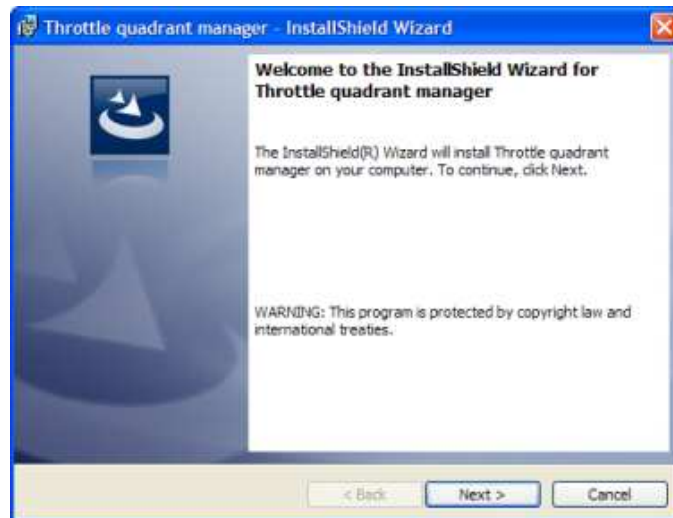
With the calibration values saved at the Factory, there is no need for you to calibrate in Windows or use any third party module such as FSUIPC.

In the unlikely event that you need to Reset any axis, we have included in the software package a Calibration Utility.

If you do decide that re-calibration is required, please follow the procedures laid out in this manual.

## SOFTWARE INSTALLATION

Once you have downloaded the Installation program, run SETUP to start the installation.



Please follow the on-screen instructions to install the program. When given the option to choose a destination folder, we recommend that you accept the default so that the Revolution Simproducts programming is placed in your Program Files Folder.



Révolution Simproducts

Once the installation is complete, navigate to the Revolution Simproducts folder that will have been installed in your Program Files Folder. In the Revolution Simproducts folder, you will see the Throttle Quadrant Manager has been installed. Please open this folder and you will see the different execute files included.



R-sim throttle quadrant  
Calibration.exe  
R-sim throttle quadrant

Your new TQ will have been 'Factory Calibrated' and the values stored. But in the unlikely event that you encounter a problem with any axis, the software



RSim throttle reset card.exe  
RSim throttle sans moteur

installation includes a utility to allow re-calibration of any axis.

We suggest that you create a shortcut onto your desktop for both the Calibration and the Reset Card .exe files at this time.

Calibration is done in the RevSim Card Calibration Pages. There is no need to calibrate in Windows. Everything is taken care of in our Calibration Manager. There is also no need to use third party modules such as FSUIPC.

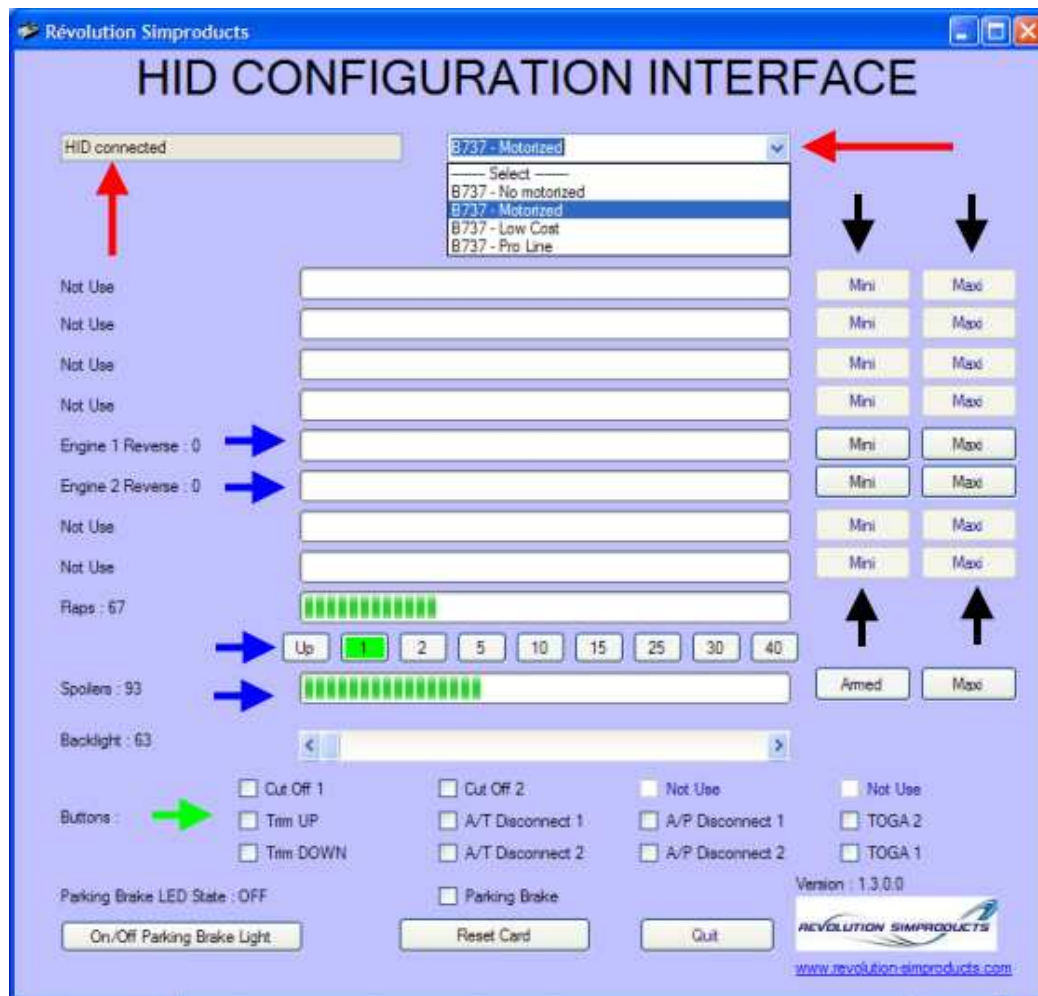
Before calibrating your card, please make sure that you have a 12vdc supply to your TQ and that the card's USB lead is attached to a spare USB socket on your PC.

Once everything is connected, you may now open the R-sim throttle quadrant Calibration.exe file and the HID Configuration Interface Page will open.

On this page you will be able to check operation of the axes and buttons before use and re-calibrate any axis should the need arise.

IMPORTANT: Version 1.6 TQ owners whether it is PROLINE or Motorized MUST choose Motorized in the top screen.

Version 1.7 TQ owners whether it is PROLINE or Motorized MUST choose PROLINE in the top screen.



### **UNDERSTANDING THE CONFIGURATION PAGE**

If all connections are good and the card is being seen, then in the top left box, you will see the message 'HID connected'.

In the right hand box, please select from the dropdown menu which type of TQ you have attached. In our example we have chosen the B737 – Motorised. When you make your choice here, the axes you need for calibration are automatically loaded.

**If you are unsure as to which TQ to select, Please Contact Us**

The centre boxes contain the axis indicator boxes. As you move an axis, it's value will be displayed in the corresponding indicator box as shown here & the Flaps and Spoiler Boxes.



Below the indicator boxes are the Button Indicators. When a button is pressed, a tick will appear in the corresponding box and when the button is released the box will return to an empty state.

At the bottom of the page you may check the Parking Brake Indicator, you may reset the card to a default state or you may Quit the Configuration Program altogether.

And finally on the right side of the page are the Mini and Maxi buttons where you set the range of each axis we are using.

## **FUNCTIONS**

Here's the table showing what functions are used on the Revolution Simproducts Controller for each type of Throttle Quadrant. Functions used with our card are coloured Yellow. So for example, a Non Motorised TQ uses the Throttle 1 & 2, Reverser 1 & 2, Spoiler & the Flaps Input onto the card. Or, a Motorised 1 TQ will use the Reversers, Flaps & Spoiler etc.

**If you are unsure which type of TQ that you need to select, please contact us.**

|              | Gaz 1 | Gaz 2 | Gaz 3 | Gaz 4 | Reverse 1 | Reverse 2 | Reverse 3 | reverse 4 | spoiler | flaps |
|--------------|-------|-------|-------|-------|-----------|-----------|-----------|-----------|---------|-------|
| No motorized |       |       |       |       |           |           |           |           |         |       |
| Motorised 1  |       |       |       |       |           |           |           |           |         |       |
| Motorised 2  |       |       |       |       |           |           |           |           |         |       |
| Pro Line     |       |       |       |       |           |           |           |           |         |       |

## **CHECK THE BUTTON INPUTS**

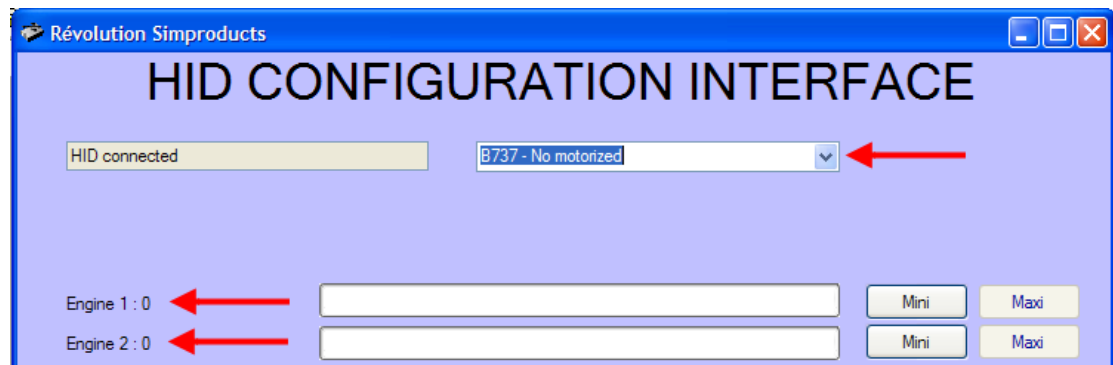
First, please check that all button inputs are being seen, a total of 10 buttons. When a button is pressed, a tick will appear in the corresponding box and when it is released, the box will return to an empty state.

Please remember for the card to see the Trim Up and Trim Down buttons, you will need to turn the left hand switch in the hardware STAB TRIM panel into it's down position.

### **CALIBRATING THE AXES**

Customers with a Non-Motorised TQ will have to calibrate both throttle inputs in addition to the Reversers, Spoiler & Flaps. Customers with any version of the Motorised TQ can skip this part and go straight to the section on the REVERSERS.

#### **For Customers With a Non-Motorised TQ Only**

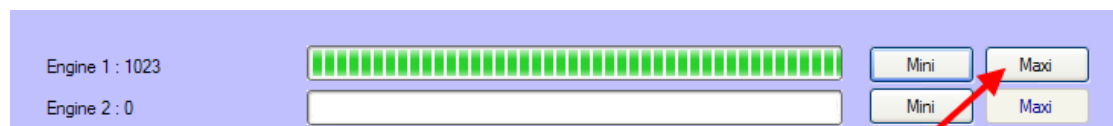


When you make your TQ selection in the HID Interface dropdown menu, you will select B737 No Motorised. Doing this will make the Engine 1 and Engine 2 Throttle calibration routines available.

To Calibrate a Throttle Lever potentiometer (in our example Throttle 1), move the lever to it's idle position and click the Mini button.



Move the lever to it's other extreme. You will see the value in the indicator box has increased. Now click the Maxi button to record the range of the potentiometer. The Engine 2 Throttle calibrates in exactly the same way.



**REVERSER 1.** Raise the Reverse Lever Knob about 15mm from it's down position. This to ensure we return to a Zero Value when the Lever is down. Click the Mini button alongside the Reverser 1 Indicator Box. See also the note at the end of this document on Reversers.

|                      |                      |      |      |
|----------------------|----------------------|------|------|
| Not Use              | <input type="text"/> | Mini | Maxi |
| Engine 1 Reverse : 0 | <input type="text"/> | Mini | Maxi |
| Engine 2 Reverse : 0 | <input type="text"/> | Mini | Maxi |
| Not Use              | <input type="text"/> | Mini | Maxi |

Move the lever to it's fully up position. You will see that the indicator bar has now filled the box. Now click the Maxi button alongside the Reverser 1 Indicator box.

|                        |                        |      |      |
|------------------------|------------------------|------|------|
| Not Use                | <input type="text"/>   | Mini | Maxi |
| Engine 1 Reverse : 253 | <div><div></div></div> | Mini | Maxi |
| Engine 2 Reverse : 0   | <input type="text"/>   | Mini | Maxi |
| Not Use                | <input type="text"/>   | Mini | Maxi |

That's the Reverser 1 calibrated, now let's do the Reverser 2.

**REVERSER 2.** Raise the Reverse Lever Knob about 15mm from it's down position, This to ensure we return to a Zero Value when the Lever is down. Click the Mini button alongside the Reverser 2 Indicator Box. See also the note at the end of this document on Reversers.

|                      |                      |      |      |
|----------------------|----------------------|------|------|
| Not Use              | <input type="text"/> | Mini | Maxi |
| Engine 1 Reverse : 0 | <input type="text"/> | Mini | Maxi |
| Engine 2 Reverse : 0 | <input type="text"/> | Mini | Maxi |
| Not Use              | <input type="text"/> | Mini | Maxi |

Move the lever to it's fully up position. You will see that the indicator bar has now filled the box. Now click the Maxi button alongside the Reverser 2 Indicator box.

|                        |                        |      |      |
|------------------------|------------------------|------|------|
| Not Use                | <input type="text"/>   | Mini | Maxi |
| Engine 1 Reverse : 0   | <input type="text"/>   | Mini | Maxi |
| Engine 2 Reverse : 255 | <div><div></div></div> | Mini | Maxi |
| Not Use                | <input type="text"/>   | Mini | Maxi |

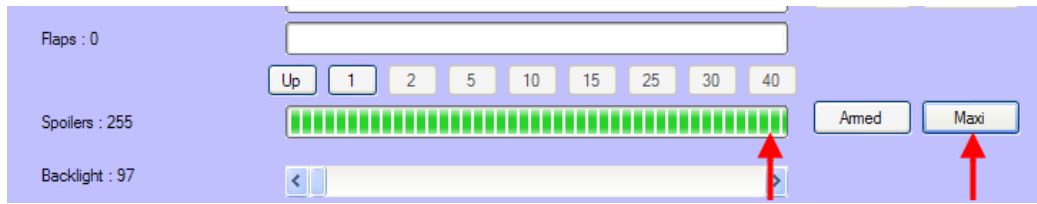
Ok, that's Reverser 2 calibrated, now let's do the Spoiler

**SPOILER.** Place the Spoiler Lever in the Armed position. You will see that the value in the indicator box is showing a partial fill of the box. Click the button in the Configuration Interface window with the legend Armed. Once the Armed position has been registered, the Armed Button will turn green.

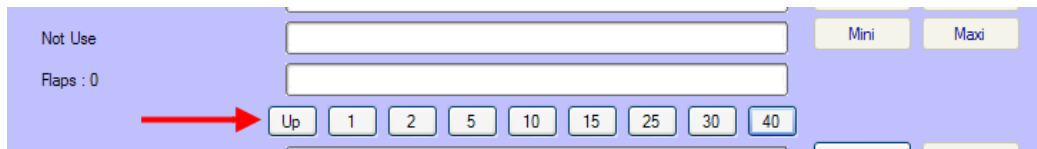
|                |                         |       |      |
|----------------|-------------------------|-------|------|
| Flaps : 0      | <input type="text"/>    |       |      |
|                | Up 1 2 5 10 15 25 30 40 |       |      |
| Spoilers : 48  | <div><div></div></div>  | Armed | Maxi |
| Backlight : 64 | <input type="text"/>    |       |      |

Now move the Spoiler Lever to it fully up position. See the indicator bar has now filled the box. Click the Maxi button to register the fully up position.

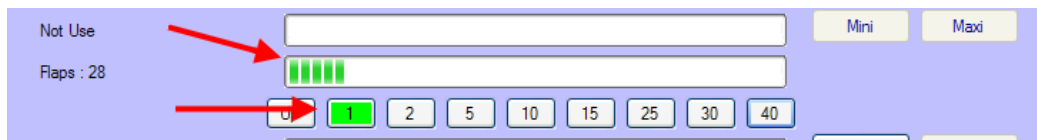
**IMPORTANT: IF YOUR TQ IS V1.7 YOU DO NOT NEED TO CALIBRATE REVERSERS...SKIP THAT STEP**



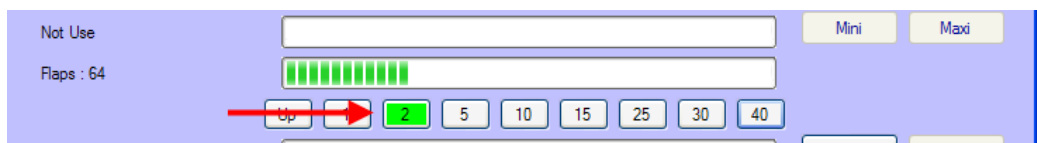
**FLAPS.** Place the Flaps Lever in the UP position. Click the button in the Configuration Interface window in the Flaps section that says Up.



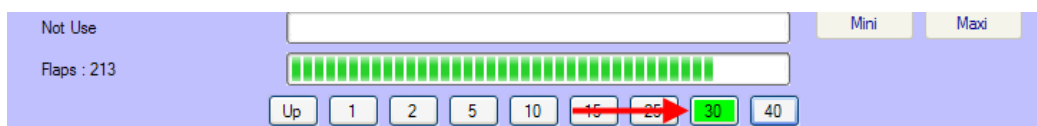
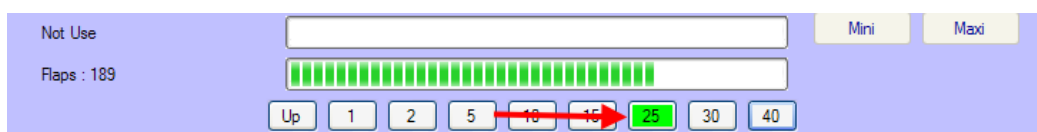
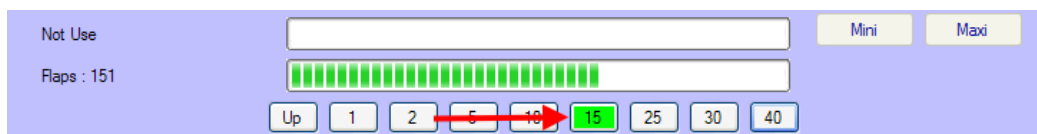
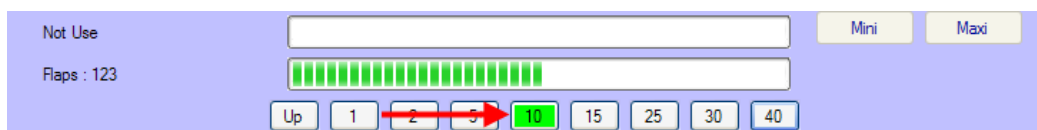
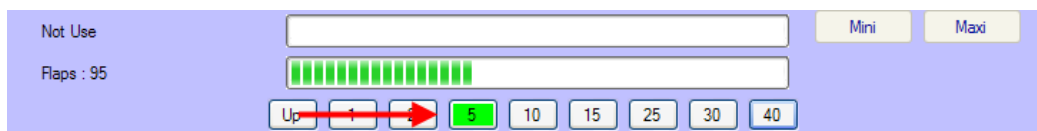
Move the Lever to the Flaps 1 position. You will see a value registered by a partial fill of the indicator box. This time click the button that says 1.

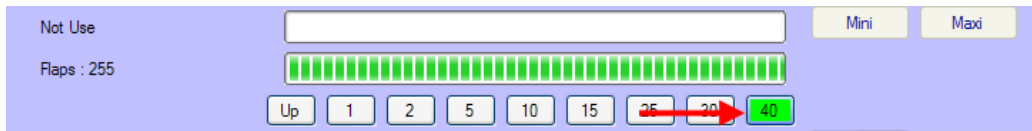


Move the lever to the Flaps 2 position, now click the button that says 2.



Repeat for each Flaps position always clicking the button that has the same value as the Flaps position selected.





### **CHECKING YOUR CALIBRATION**

Once calibration is complete, you may now check operation of the axes by looking at the values in the indicator boxes as you move the axis.

As you move the Spoiler axis, when passing the position registered as the Armed position, the Armed button will flash Green.

As you move the Flaps axis, when passing each position registered as a Flaps position, the corresponding Flaps button will flash Green.

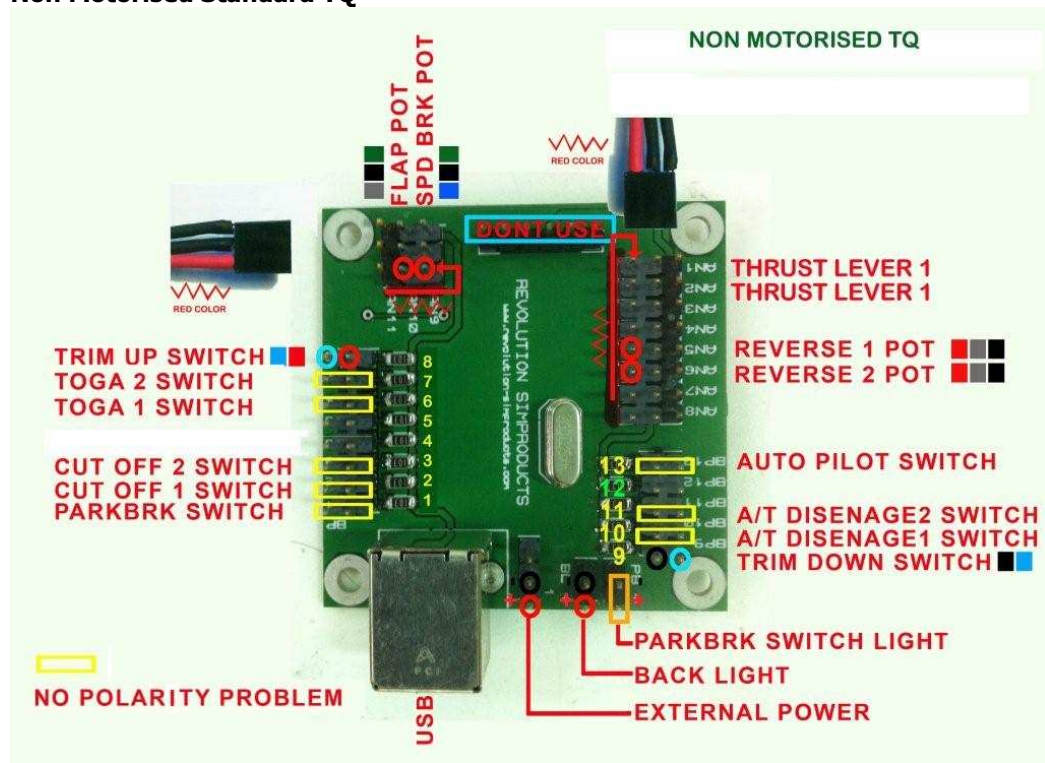
### **ERRORS.**

It is important to follow the sequence of calibration settings as above. It is only possible to set each axis calibration once. Attempting to set, then reset an axis calibration may result in an Error Message.

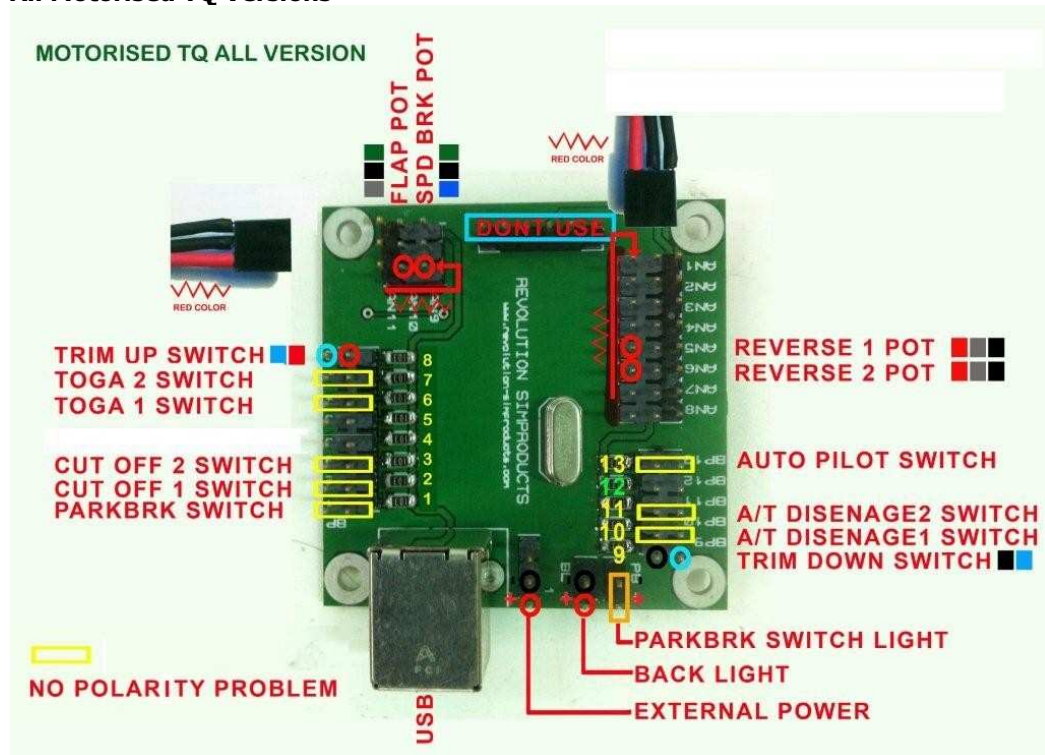
Should you receive an Error Message, you will need to Reset the Card to a Default State and start again with the calibration.

## CONNECTIONS TO THE CARD

### Non Motorised Standard TQ



### All Motorised TQ Versions



Above Picture is only for V1.6 TQ's.

V1.7 TQ's do not connect the reversers to this card.

**12vdc Input** The 12vdc Input (External Power) to the Card is Polarity Sensitive. The PCB is marked which pin is + (positive) shown in the diagram with the **red circle**.

#### **NOTES:**

**Backlight Output** The Backlight Output which is marked BL is again Polarity Sensitive. The PCB is marked which pin is + (positive) shown in the diagram with the **red circle**.

**Parking Brake Light** The Parking Brake Light Output which is marked PB has a + & - output. However, the Parking Brake Lamp is NOT Polarity sensitive, so it is not important which way round it is connected.

**Switch Connections** All Switch Connections are by way of 2 way header pins having standard 2.54mm pitch onto the Card. Each switch input has a number prefixed by the letters BP. Switches are connected to the inputs shown in the above diagram. It is not important which way round the switch wires are connected to the card as these are sensing continuity and not polarity. Marked on the diagram with a yellow box.

PARK BRAKE SWITCH = BP1  
CUTOFF 1 SWITCH = BP2  
CUTOFF 2 SWITCH = BP3  
TOGA 1 SWITCH = BP6  
TOGA 2 SWITCH = BP7  
TRIM UP SWITCH = BP8  
TRIM DOWN SWITCH = BP9  
A/T DISENGAGE 1 SWITCH = BP10  
A/T DISENGAGE 2 SWITCH = BP11  
AUTOPILOT SWITCH = BP13

**Note On Trim Up/Down Switches.** The two trim switches share a common GND connection. You will notice that on the diagram, the two inputs have different colours yet they are accompanied by a common coloured GND connection.

**AXES Inputs** All axes inputs to the card are by way of 3 way header pins again having a standard pitch of 2.54mm. All axes inputs have a letter prefix of AN. It is important that you understand which pin should be connected to the Wiper (Variable) on the potentiometer. ALL VARIABLE INPUTS are the inside pin which faces the centre of the card as you will see from the diagram. The variable inputs are marked on the diagram with a **RED CIRCLE** and a **RED SLEEVE** on the Plug. The remaining pins are for the + & GND connections to the potentiometer.

THRUST LEVER 1 = AN1           **\*\* For Non Motorised Only**  
THRUST LEVER 2 = AN2           **\*\* For Non Motorised Only**

#### **All Motorised TQ Thrust Lever Potentiometers are connected to the second card**

REVERSE 1 = AN5  
REVERSE 2 = AN6  
SPOILER = AN10  
FLAPS = AN11

#### **REVERSERS**

Switching back from Reverse Thrust to Normal Idle occurs at only one point on the Reverse Potentiometer when it's value is absolute zero. All potentiometers have a +/- tolerance, so to ensure we return to a zero value when the Reverse Lever is in it's down position, we set the Mini point with the pot slightly advanced on it's travel, hence raising the lever slightly to set the Mini value.

## Hints & Tips :

**BrakeReleaseThreshold=75 line.** To have this function available, you must update to the latest version of FSUIPC if you have not already done so. We appreciate that you may have already set up several other inputs through FSUIPC like buttons and switches, macro's, axes etc. This information has already been saved in the FSUIPC.ini file. So, we recommend that you simply edit the file using Notepad. First, check to see if the line BrakeReleaseThreshold=75 appears in the section under [General]. If not, simply add the line in the [General] section of the file and save it.

**Beware of 'Cross Controls'.** That is where a key or other input is assigned in both FS (or FSX) and also in your Add On Aircraft or Avionics Package.

We recommend if possible that all Lite Version TQ Owners assign **all their axes in FSUIPC** as we find it more accurate and reliable than the internal Flight Simulator Axes Control. It is also far more accurate. In the Menu Items, you can Disable Joysticks. This will not affect Button Inputs, which will still be seen.

**Use of USB Powered Hubs.** There never seems to be enough USB slots on our computers and we all want to 'MAX OUT' our input devices. But, there is only so much that your PC can tolerate. Using Powered USB Hubs seems to be one solution. But even when using Powered Hubs, your PC is still sharing the 5v bus, IRQ and I/O resources to slots that have the Powered Hub connected too. Over working the USB Bus can voltage drop and loss of configuration. If you experience bad things happening with USB devices (even when a Powered Hub is used), we recommend fitting a higher capacity PC Power Supply. Most standard PC's are built using a 400 or 450 watt PSU. Upgrading to a 650 or 750 watt PSU should cure your USB Device issues.

## Troubleshooting :

### *The throttle levers and/or trim wheels, do not move.*

- Make sure you have connected the TQ Power Supply to the electricity supply.
- Check that SIOC is running.
- Make sure Auto Pilot and Stab Trim Cutout switches are open.
- Make sure the IDX & Device numbers you have added to SIOC.ini file matches those shown in the Main SIOC Program Window.
- Press the reload button on the SIOC main screen.
- Contact us at [support@revolution-simproducts.com](mailto:support@revolution-simproducts.com)

### *Flaps or spoiler do not move in the simulator even though I change it on the TQ.*

- Ensure that your second USB connection is connected to the PC (PROLINE & Custom TQ's) Make sure you have done all the assignments correctly as shown in this manual.
- Make sure that there is no interference on the commands assigned with other joysticks or keyboard shortcuts in the main flight simulator assignments window. Make sure that the PC recognizes the joystick card (Lite Version)
- Contact us at [support@revolution-simproducts.com](mailto:support@revolution-simproducts.com)

### *Parking brakes are operating in a reverse action i.e. I release it and it sets.*

- You have assigned the Parking Brake Button as 'Parking Brake'. This is purely a TOGGLE action that changes the state of the Parking Brake. That is, if it is On, it will turn Off and vice-versa. Make sure the hardware parking brake position matches the initial position with the simulator at start of flight. Or, you may wish to use the Offset Byte Set function in FSUIPC to accurately write the correct parameter value to Offset 0BC8 (Parking Brake, 0=Off & 1=On).

### *PROLINE USERS - My Trim Wheels Do Not Turn Automatically*

The Trim Motors are wired through the Main Elec Switch on the Stab Trim Panel. This is to simulate the real action of the trim wheels. So, in the Cutoff Position, they will not work. Please ensure that your Main Elec switch is NOT set in the Cutoff Position and is set ON in the UP position.

**We are updating our web site and FAQ section frequently. Please do not forget to check our web site for the problems you are facing. As this might already been solved and published in our FAQ section.**